

Calorie Estimation using Ensemble Technique from Recipes of Bangladeshi Foods

Md. Siam Ansary

Department of Computer Science and Engineering
Ahsanullah University of Science and Technology
Dhaka, Bangladesh
ORCID: 0000-0002-4942-2541

Amina Brinto

Department of Obstetrics and Gynaecology
Kurmitola General Hospital
Dhaka, Bangladesh
{aminabrinto, brintoamina}@gmail.com

Abstract—Chingri malai curry, Ilish macher paturi, Patishapta pitha - these are well known foods of Bangladesh that we have all grown up with. Culturally rich Bangladesh has lots of foods native to it. The recipes of such food items are passed down from generation to generation and gradually they have gained perfection. In this research, we have focused on calorie estimation of these Bangladeshi foods. A dataset containing recipes of Bangladeshi foods has been created and using an BERT based stacking ensemble model, the calories have been predicted. We have observed that the ensemble model can do prediction with great efficiency using the text based dataset. The dataset, codes and all relevant resources of the experiment will be made public for reproduction and verification purposes.

Index Terms—ensemble learning, machine learning, nlp, dataset, regression

I. INTRODUCTION

Bangladesh has a culturally rich and diverse food heritage, shaped by its geography, history, and multicultural influences. Bangladeshi cuisine is known for its bold flavors, aromatic spices, and deep cultural ties to daily life, religion, and festivals. Each region of Bangladesh offers unique dishes and preparations. While Dhaka is famous for its rich Mughlai-inspired dishes like Kacchi Biryani and Bakarkhani, Chittagong offers spicier fare, including Mezban beef curry. The cuisine reflects a blend of Mughal, Persian, Arab, and British colonial influences, along with traditional Bengali roots. Special dishes are often prepared for religious occasions like Eid, Pohela Boishakh, and Durga Puja. The country has an incredibly diverse and flavorful culinary culture that reflects its rich history, regional identities, and deep-rooted traditions.

Research on calorie estimation from food recipes has become important for several compelling reasons, especially in the context of health, technology, and food science. Accurate calorie estimation can help individuals make informed dietary choices and manage calorie intake, contributing to healthier lifestyles. People with specific health conditions need precise control over calorie and nutrient intake and calorie estimation tools enable custom meal planning based on individual nutritional requirements. Unlike packaged food, home-cooked and culturally diverse meals often lack labeled nutritional information. Research in this area ensures broader applicability, capturing calorie data across different cuisines and preparation

styles. Henceforth, we have decided to conduct research on calorie estimation from recipes of Bangladeshi foods.

We have used a BERT based ensemble model on the dataset that uses bert-base, roberta-base and distilbert-base. Predictions from these have been used as features to train a second-level model, meta-regressor for which we have used Ridge Regression.

In the following sections, we have discussed relevant existing studies, our novel dataset and application of the ensemble model along with evaluated results.

II. LITERATURE REVIEW

There have been a number of research works on food calorie prediction. We have discussed the studies below. While these works have been insightful, we have observed that in most of these works, researchers have tried calorie estimation from images. Such approach may be feasible when large amount of well photographed food pictures are available of items that are globally very popular. But we believe that rather than calorie estimation from images, estimation from recipe texts is more realistic for Bengali native foods. In Bangladesh, recipes are passed down through generations and there are many regional and tribal foods whose well photographed photos are not abundantly available in large volumes.

Ma et al. [1] proposed an encoder-decoder based framework that aimed at improving the accuracy of food energy estimation from single food images. Their experimental method employed a multi-task learning approach that simultaneously predicts food segmentation, food category, and energy density. As a result, capturing spatial and semantic inter-dependencies are more effective than previous research models. A key factor was the integration of a self-attention mechanism within the decoder, which refined spatial feature localization and improved performance in complex visual scenes with overlapping food items. Additionally, they also introduced a cross-task consistency loss which aligned segmentation outputs with energy predictions, thus, reducing estimation errors.

Shao et al. [2] presented an end-to-end framework for food portion estimation which is able to leverage 3D shape reconstruction from a single monocular image, thus, addressing a central challenge in visual dietary assessment. Unlike the traditional methods that relied on multi-view images or

reference objects, this experimental approach reconstructed food geometry using a novel deep learning model trained to infer volumetric shape directly from RGB inputs. The framework integrated depth estimation and shape recovery in a unified pipeline, improving estimation accuracy without additional hardware requirements.

In order to improve estimation accuracy from monocular images, Vinod et al. [3] presented a novel image-based framework for food energy estimation. This model includes depth domain adaptation. The researchers presented a two-step method that used adversarial training to adapt the depth maps they had initially predicted from food photos to a standardized depth domain. This experimental approach improved generalization across a range of imaging conditions as the method reduced disparities between synthetic and real-world data. In order to ensure consistency and boost overall performance, the model was further improved with a multi-task learning strategy. The strategy jointly estimated segmentation, depth, and energy.

For enhancing the performance in multimodal dietary analysis, Yao et al. [4] worked on CaLoRAify. The method is an efficient and innovative framework for calorie estimation. The approach can combine visual-text pairing with Low-Rank Adaptation driven visual language models. The researchers used paired food photos and textual descriptions to enhance semantic understanding and increase estimation accuracy through building on recent developments in vision-language models (VLMs). One significant advancement was the application of LoRA to efficiently adapt to the calorie estimation task through fine-tuning large pre-trained VLMs with little computational overhead. Robust inference was made possible even in visually ambiguous food scenarios. The model's use of cross-modal attention mechanisms to be able to match textual cues with visual features helped in the regard.

In order to identify food items and determine their caloric content, Patil et al. [5] presented a calorie estimation system. The method combines conventional image processing techniques with object detection using YOLOv4. Taking advantage of the YOLOv4 model's speed and accuracy in object localisation, the research method first used it to detect multiple food items in a single image in real-time and with accuracy. A database containing caloric values was utilized in conjunction with segmentation and area estimation techniques, so that processing can be done to detect items and determine approximate calorie values. Under certain conditions, the scientists showed that the system could process different kinds of food and provide estimates that were reasonably accurate.

Ayon et al. [6] presented FoodieCal with the target of automating dietary analysis of food images. The experimental study is a food detection and calorie estimation system. The method used convolutional neural networks (CNNs). Using a trained CNN model, the system was developed to recognize different food items in an image. It then mapped the recognized classes to a caloric data to estimate the food items' caloric content. Deployability was prioritized for the experimental method. Also, focus was given on practical uses

like dietary tracking tools and smartphone applications.

Convolutional neural networks (CNNs) were used in the experimental study by Makwana et al [7]. They analyzed food photos and calculated calorie content in their food recognition and calorie assessment system. The food items were identified from photos through training a CNN model. And then, the identified items were mapped to a database containing nutritional values to estimate calorie. The research demonstrated that deep learning can handle visual complexity and variation in food appearances very well. The approach allowed the system to identify a variety of food types with a consistent level of accuracy.

An ingredient based food calorie estimation system was presented by Turmchokksam and Chamnongthai [8] in their research work. The method combined nutritional knowledge with image feature analysis. Brightness and thermal heat information had been utilized to improve the estimation accuracy. The system detected food ingredients and calculated calories by examining both optical brightness and thermal distribution patterns. The method is suggestive of food composition and density, in contrast to conventional methods. To enable a rule-based estimation procedure, the system integrated a nutritional knowledge base to match identified ingredients with corresponding caloric values.

Liang and Li [9] presented a comprehensive study on computer vision-based food calorie estimation. They introduced a dedicated dataset along with a baseline method for estimating caloric content from food images. Their approach combined food recognition and volume estimation to compute calories. The approach used image segmentation and geometric assumptions to approximate portion sizes. The study emphasized the importance of high-quality datasets and outlines key challenges such as food variability, occlusion and lighting conditions.

Yanai and Kawano [10] proposed a food image recognition system using deep convolutional neural networks (CNNs). The method was enhanced by pretraining on large-scale datasets followed by fine-tuning on food-specific images. The experimental approach addressed the challenge of food classification, which involves a high intraclass variation and visual similarity between different dishes. By making use of transfer learning, the model achieved improved accuracy even with limited labeled food data. The study demonstrates that fine-tuning pretrained CNN models can significantly improve the performance over training from scratch. Thus, the method was deemed both effective and efficient for food recognition tasks.

Pouladzadeh et al. [11] presented a practical system for food calorie measurement. The research combined image processing and computer vision techniques to estimate caloric intake from meal images. The system involved detecting and segmenting food items from images through estimation of portion sizes. This was done using volume approximation methods, and mapping identified foods to a nutritional database for calorie computation. The method emphasized user-friendliness. The approach also aimed to assist individuals in managing their diet more accurately without relying on

manual input.

Anthimopoulos et al. [12] developed a food recognition system. The researchers identified the need of such research for diabetic patients. Bag of Features (BoF) had been used for the carbohydrate estimation technique. The system aimed to assist diabetics in managing diet by automatically identifying food items from images and estimating nutritional content. The content estimation was particularly on carbohydrate levels. The research was able to enhance the traditional BoF approach by integrating color and texture features and applying a support vector machine (SVM) for classification.

Bossard et al. [13] introduced the Food-101 dataset which is a large-scale benchmark for food recognition. The scientists proposed a novel approach that combines random forests with discriminative patch mining, thus, improving food classification performance.

Kawano and Yanai [14] addressed the challenge of limited annotated data in food recognition. They proposed a method to automatically expand food datasets. Their approach utilized existing image categories and applies domain adaptation techniques. Thus the study bridges the gap between web-sourced images and real-world food photographs. By refining the dataset with domain-adapted classifiers, the researchers enhanced the quality and relevance of newly added images while minimizing noise.

Fard et al. [15] presented a system that leveraged convolutional neural networks (CNNs) to recognize various fruits and vegetables along with estimating their caloric content. The proposed method involves classifying food items through a CNN-based model and calculating calories based on recognized food types and predefined nutritional values. The system aimed to assist users in managing dietary intake more effectively through automating the calorie estimation process.

Yanai and Kawano [16] worked on a food recognition system. It was designed for mobile devices, allowing users to identify foods in real time scenarios. The system utilized a combination of feature extraction methods and classification algorithms that are optimized for computational efficiency on mobile platforms. By focusing on fast processing and real-time performance, the researchers addressed the challenges of limited hardware resources and the need for quick responses in mobile applications.

III. THE NOVEL DATASET

We have developed a novel and comprehensive dataset of recipes of different Bangladeshi food items. The items are native to Bangladesh. The recipes are in English. The dataset has been prepared as a CSV file that has three columns which are

- Food Item Name (text)
- Detailed Recipe (text)
- Approximate Calorie Value (numeric)

We have collected the recipes of foods of different regions and cultures of the country. The detailed recipe text of each item describes the ingredients and their precise amounts, as well as the preparation steps.

The calories per serving of the food items have been estimated, examined, and verified by multiple nutrition experts and dietitians.

IV. APPLICATION OF THE ENSEMBLE APPROACH

Our dataset has three columns as discussed in the previous section. We have dropped the *Food Item Name* and have used the *Detailed Recipe* column for estimating the values of *Approximate Calorie Value*. The dataset is partitioned into three portions: train (seventy%), test (fifteen%) and validation (fifteen%).

We have used a stacking ensemble approach in our study. At first, we have used bert-base-uncased, roberta-base and distilbert-base-uncased for predicting calorie values and then, these values are used as features for the meta regressor, a Ridge regression model.

While training the transformer-based models, for each model, we at first have tokenized recipe text for each data split. For ensuring that the inputs fit for models, maximum length was set to five hundred and twelve. The encoded inputs were wrapped and labeled into PyTorch compatible datasets. A batch size of sixteen is used during training and evaluation. Each model was trained for ten epochs. The initial learning rate was set to $2e-5$. The low learning rate is able to help prevent forgetting of pretrained knowledge. Weight decay was set as 0.01 to prevent overfitting by penalizing large weights.

As we have done stacking ensemble, predictions from the transformer models were combined. On validation predictions and labels, a Ridge regression model was fit. This model was used for the final prediction.

V. EVALUATED RESULTS

Since the task at hand is a regression problem, we have used the following metrics for evaluation purposes.

- Mean Squared Error (MSE): It is the metric for calculating the average value of the squares of the differences between actual and predicted values.
- Root Mean Squared Error (RMSE): This is the evaluation metric for square root of MSE.
- Mean Absolute Error (MAE): It is the metric for calculating the average of the absolute differences between the actual and the predicted values.
- Coefficient of Determination (R^2 Score): It is the metric for measuring how well the regression model explains the variance in the target, ranging from 0 (no fit) to 1 (perfect fit).
- Mean Absolute Percentage Error (MAPE): This evaluation metric is for calculating the average percentage difference between the actual and the predicted values.

The evaluated results of the study are presented on the corresponding GitHub Repository [17].

The stacking ensemble regression model has achieved excellent results for estimating calories from the recipes. The approach has effectively combined the transformer-based models. Each model has been able to capture different semantic and contextual nuances from the recipe descriptions, and

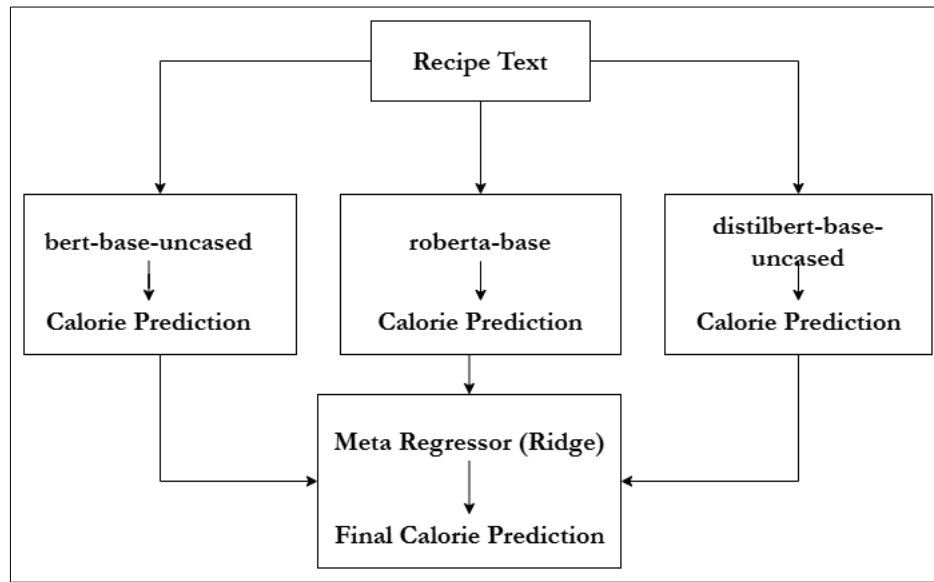


Fig. 1: The steps of the ensemble approach

their diversity have helped reduce both bias and variance in the predictions. The Ridge regression meta-regressor further has been able to enhance performance by learning optimal weights for combining base model outputs while applying regularization to prevent overfitting. This combined approach has allowed the ensemble to generalize well across a wide range of recipes. As a result, the model has been able to perform excellently.

VI. THE AVAILABILITY OF THE RESOURCES

The resources related to the study are made publicly available on a GitHub repository by the researchers. This is done so that more researchers can utilize the dataset and do more work on Bangladeshi Foods. The corresponding GitHub repository is MdSiamAnsary/Calorie-Prediction [17].

VII. CONCLUSION AND FUTURE WORKS

Bangladeshi people love to eat. Nowadays there are people in Bangladesh who have become professional food vloggers. But, not many Bangladeshi people are health conscious and as a result, they tend to suffer from various diseases and health complications. It would be great if Bangladeshis can get a sense of their calorie intakes throughout their love for food. It can help them in great manner. Through this research, a dataset has been developed for calorie prediction from recipes of Bangladeshi foods. The BERT based ensemble model has yielded great results while doing the predictions. In the future, we hope to enrich the dataset with more records and experiment with more models to improve efficiency. Also, we will develop another dataset where the recipes will be in the native language of Bangladesh. An application development can also be done so that the research can have practical usage.

ACKNOWLEDGMENTS

The researchers, Md. Siam Ansary and Amina Brinto, siblings, are grateful to their parents, Dr. Mohammad Shahrier Ansary and Mst. Afroza Khanom for their continuous support.

The researchers are also thankful to the individuals helping them in creation of the dataset and its verification process. The contribution of the open source community has also helped a great deal in the implementation process.

REFERENCES

- [1] J. Ma, J. He, and F. Zhu, "An improved encoder-decoder framework for food energy estimation," *arXiv preprint arXiv:2309.00468*, 2023.
- [2] Z. Shao, G. Vinod, J. He, and F. Zhu, "An end-to-end food portion estimation framework based on shape reconstruction from monocular image," *arXiv preprint arXiv:2308.01810*, 2023.
- [3] G. Vinod, Z. Shao, and F. Zhu, "Image based food energy estimation with depth domain adaptation," *arXiv preprint arXiv:2208.12153*, 2022.
- [4] D. Yao, K. Yao, J. Zhou, and Y. Zhang, "CaLoRAify: Calorie estimation with visual-text pairing and LoRA-driven visual language models," *arXiv preprint arXiv:2412.09936*, 2024.
- [5] S. Patil, S. Patil, V. Kale, and M. Bonde, "Food item calorie estimation using YOLOv4 and image processing," *Int. J. Comput. Trends Technol.*, vol. 69, no. 5, pp. 69–76, 2021.
- [6] S. A. Ayon, C. Z. Mashrafi, A. B. Yousuf, and F. Hossain, "FoodieCal: A convolutional neural network based food detection and calorie estimation system," *Natl. Comput. Coll. Conf. (NCCC)*, 2021.
- [7] Y. Makwana, S. Iyer, and S. Tiwari, "Food recognition and calorie assessment from images using convolutional neural networks," *Int. J. Intell. Syst. Appl. Eng.*
- [8] S. Turmchokksam and K. Chamnongthai, "The design and implementation of an ingredient-based food calorie estimation system using nutrition knowledge and fusion of brightness and heat information," *IEEE Access*, vol. 6, pp. 28370–28378, 2018.
- [9] Y. Liang and J. Li, "Computer vision-based food calorie estimation: Dataset, method, and experiment," *arXiv preprint arXiv:1705.07632*, 2017.
- [10] K. Yanai and Y. Kawano, "Food image recognition using deep convolutional network with pre-training and fine-tuning," in *Proceedings of IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, 2015.
- [11] P. Pouladzadeh, S. Shirmohammadi, and A. Yassine, "You are what you eat: So measure what you eat!," *IEEE Instrum. Meas. Mag.*, vol. 19, no. 1, pp. 40–47, 2016.

- [12] M. Anthimopoulos, L. Gianola, L. Scarnato, P. Diem, and S. Mougiakkou, "A food recognition system for diabetic patients based on an optimized bag of features model," *IEEE J. Biomed. Health Inform.*, vol. 18, no. 4, pp. 1261–1271, July 2014.
- [13] L. Bossard, M. Guillaumin, and L. Van Gool, "Food-101 – Mining discriminative components with random forests," in *Proceedings of Eur. Conf. Comput. Vis. (ECCV)*, pp. 446–461, 2014.
- [14] Y. Kawano and K. Yanai, "Automatic expansion of a food image dataset leveraging existing categories with domain adaptation," in *Proceedings of ECCV Workshop Transferring Adapt. Source Knowl.*, 2014.
- [15] M. A. Fard, H. Hadadi, and A. T. Targhi, "Fruits and vegetables calorie counter using convolutional neural networks," in *Proceedings of 6th Int. Conf. Digital Health Conf.*, pp. 11–13, 2016.
- [16] K. Yanai and Y. Kawano, "Real-time mobile food recognition system," in *Proceedings of IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, pp. 1–7, 2013.
- [17] "MdSiamAnsary/Calorie-Prediction", [Online] Available: <https://github.com/MdSiamAnsary/Calorie-Prediction>, Last Accessed on: 19 September 2025.